SkillNavora

MODULE 10 HANDBOOK

Module 10: Enterprise Solutions Architecture and System Integration

5 sections, every topic explained with how it works, examples and resources.

From Non-Coder to Enterprise AI Builder: Learn, Build, Deploy, Get Hired and Launch Your AI Business.

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Financial figures are hypothetical examples, not advice.

Module 10: Enterprise Solutions Architecture and System Integration

Every section below explains the concept, how it works, each listed topic in detail, a worked example and where to learn more.

Sections

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2. 10.2 Enterprise Integration
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MODULE 10 · SECTION 10.1

10.1 Enterprise Architecture

DEFINITION

> Enterprise architecture describes how an organisation's applications, data, integrations and infrastructure fit together and how changes should be made.

HOW IT WORKS

- Views: application, data, integration, infrastructure and now AI architecture.
- Principles guide decisions (for example 'buy before build', 'data stays in region').
- An Architecture Decision Record (ADR) records one decision: context, options, decision, consequences.

EVERY TOPIC EXPLAINED

Application, data, integration, infrastructure and AI architecture. Application architecture: the systems and how they interact. Data architecture: where data lives, its models and flows. Integration architecture: how systems connect (APIs, events, files). Infrastructure architecture: compute, network and storage. AI architecture: models, gateways, retrieval, agents, evaluation and governance.

Architecture principles, documentation, architecture decision records. Architecture principles are agreed rules that guide decisions (security by design, API-first, reuse before build). Architecture documentation includes context, container and component diagrams (the C4 model), data flows and non-functional requirements. Architecture decision records (ADRs) capture each significant decision with context, options, choice and consequences.

WORKED EXAMPLE

> ADR-012: Use a model gateway for all LLM calls. Context: three teams call providers directly. Decision: one gateway with logging and cost limits.

RESOURCES

- ADR GitHub organisation https://adr.github.io/
- The Open Group: TOGAF https://www.opengroup.org/togaf

10.2 Enterprise Integration

DEFINITION

> Enterprise integration connects an AI solution to existing systems such as ERP, CRM, HRMS, payments and document stores, including old legacy systems.

HOW IT WORKS

- Patterns: REST APIs for request-reply; webhooks for push events; message queues (Kafka, RabbitMQ, SQS) for reliable asynchronous work.
- Middleware or an enterprise service bus translates formats between systems; an API gateway secures and meters access.
- Legacy systems without APIs may need database views, file drops or robotic process automation.

EVERY TOPIC EXPLAINED

ERP, CRM, HRMS, payment and document management integration. ERP (SAP, Oracle, Dynamics) runs finance, supply chain and operations. CRM (Salesforce, HubSpot, Zoho) manages customers and sales. HRMS (Workday, SAP SuccessFactors) manages employees. Payment systems (Stripe, Razorpay) process payments. Document management systems (SharePoint, Google Drive) store files. Each has APIs, permissions and data rules.

Legacy systems; REST APIs; webhooks; message queues; event-driven integration. Legacy systems are old but critical, often with limited APIs: integrate through database views, file exports, SOAP services or RPA. REST APIs suit request-response. Webhooks push events in real time. Message queues (RabbitMQ, SQS, Kafka) decouple systems and absorb load. Event-driven integration reacts to business events (order placed, invoice received).

API gateways, middleware, enterprise service buses. API gateways secure and route API traffic. Middleware transforms and routes messages between systems (MuleSoft, Boomi, Azure Logic Apps). An enterprise service bus (ESB) is a central integration backbone in older architectures; modern designs prefer lighter API and event-based integration.

WORKED EXAMPLE

> When a deal closes in Salesforce, a webhook posts to a queue; a worker creates the customer in SAP and the AI onboarding assistant gets a welcome task.

RESOURCES

- Enterprise Integration Patterns https://www.enterpriseintegrationpatterns.com/
- Salesforce developer docs https://developer.salesforce.com/docs

10.3 Enterprise AI Architecture

DEFINITION

> Enterprise AI architecture is the shared platform that lets many teams use AI safely: model access, retrieval, agents, governance and observability.

HOW IT WORKS

- An AI gateway sits between apps and model providers: authentication, routing, logging, cost limits, PII redaction.
- A model abstraction layer lets apps switch providers without code changes.
- Enterprise RAG respects document permissions: users only retrieve what they may read.
- Multi-tenant platforms isolate each business unit's data, prompts and logs.

EVERY TOPIC EXPLAINED

Centralised and distributed AI platforms; multi-provider LLM architectures. A centralised AI platform gives all teams one governed set of models and tools; distributed means each team runs its own. Multi-provider LLM architectures use several providers for quality, cost and resilience. A model abstraction layer gives one internal interface so applications are not tied to one provider.

Model abstraction layers; enterprise RAG; agent orchestration. Enterprise RAG adds permissions, freshness, citations and scale to RAG. Agent orchestration coordinates agents, tools, approvals and state across business processes. AI gateways (LiteLLM, cloud API management, Kong) centralise keys, routing, rate limits, logging and cost control. Model routing sends each request to the right model by task, cost or availability.

AI gateways, model routing, data access controls, multi-tenant enterprise AI, AI observability. Data access controls ensure the AI only uses data the requesting user may see. Multi-tenant enterprise AI separates business units or customers. AI observability tracks quality, latency, cost, usage, errors and safety events across all AI applications.

WORKED EXAMPLE

> All teams call one internal endpoint; it routes summarisation to a low-cost model, legal analysis to a stronger model, and logs every call for audit.

RESOURCES

- Microsoft: Azure AI reference architectures https://learn.microsoft.com/azure/architecture/ai-ml/
- Google Cloud: generative AI architecture https://cloud.google.com/architecture/ai-ml

10.4 Security and Governance

DEFINITION

> Security and governance make sure AI systems handle data lawfully, behave as intended, and can be explained and audited.

HOW IT WORKS

- Classify data (public, internal, confidential, restricted) and decide which classes may go to which models.
- Retention rules say how long prompts and outputs are kept. Access control limits who sees logs.
- Governance tracks model risk, bias, explainability and hallucination risk, with human review for consequential decisions.
- Security reviews check the architecture before launch.

EVERY TOPIC EXPLAINED

Data privacy, classification, retention; responsible AI; AI governance. Data privacy protects personal data according to law (GDPR, India's DPDP Act). Data classification labels sensitivity. Data retention sets how long data, prompts and outputs are kept. Responsible AI ensures fairness, transparency and accountability. AI governance assigns ownership, policies and approval processes.

Model risk management, explainability, bias, hallucination risk, auditability. Model risk management assesses and monitors risks of each model use. Explainability shows how a result was reached (sources, factors). Bias checks outcomes across groups. Hallucination risk is reduced with grounding and review. Auditability means logs and versions let you reconstruct any decision.

Access control, regulatory awareness, enterprise security reviews. Access control enforces who can use which AI capability and data. Regulatory awareness tracks the laws and sector rules that apply (EU AI Act, RBI or SEBI guidelines in Indian finance, HIPAA in US health). Enterprise security reviews assess the solution's architecture, data flows, vendors and controls before approval.

WORKED EXAMPLE

> Restricted HR data may only be processed by a model deployed inside the company's own cloud tenant, with logs kept 90 days.

RESOURCES

- NIST AI Risk Management Framework https://www.nist.gov/itl/ai-risk-management-framework
- ISO/IEC 42001 overview https://www.iso.org/standard/81230.html

10.5 Enterprise Reliability

DEFINITION

> Reliability engineering keeps systems available and recovers quickly when they fail.

HOW IT WORKS

- Service-level objectives (SLOs) set targets such as 99.9% of requests succeed; service-level agreements (SLAs) are the promises made to customers.
- Fault tolerance: redundancy across zones, retries, circuit breakers and graceful degradation (fall back to a simpler answer).
- Backups and disaster recovery have tested restore times. Incidents get a blameless post-mortem with root cause and actions.

EVERY TOPIC EXPLAINED

Availability, reliability engineering, scalability, fault tolerance, resilience. Availability: share of time the service works (99.9% ≈ 43 minutes downtime a month). Reliability engineering designs for failure. Scalability handles growth. Fault tolerance keeps working when parts fail. Resilience recovers quickly from failures.

Disaster recovery, backup, incident management, troubleshooting, root cause analysis. Disaster recovery restores service after a major failure, with targets for recovery time (RTO) and data loss (RPO). Backups must be tested by restoring them. Incident management detects, triages, resolves and communicates. Troubleshooting narrows down causes using logs, metrics and traces. Root cause analysis finds and fixes the underlying cause.

Service-level objectives and agreements. A service-level indicator (SLI) is a measured value (successful requests %). A service-level objective (SLO) is the internal target (99.9%). A service-level agreement (SLA) is the contractual commitment to the customer, usually looser than the SLO, with penalties if missed.

WORKED EXAMPLE

> If the model provider times out, the app shows the top retrieved help articles instead of an error, keeping the service usable.

RESOURCES

- Google SRE book (free) https://sre.google/sre-book/table-of-contents/
- Atlassian: incident management https://www.atlassian.com/incident-management